

Problems

Status

Standings

Intermediary

Tags:

Time Limit: 1 s Memory Limit: 32 MB
Submission: 97 AC: 40 Score: 99.47

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Codes

AI Code Tips

Description

It is widely known that any two strangers can get to know each other through at most six other people. Now let's prove this.

In the country Intermediary Conducts Personal Communications (ICPC), there are up to n ($2 \leq n \leq 100$) ordinary people conveniently numbered from 0 to $n-1$. They don't know each other, or, in other words, they are strangers. The only way they can communicate with each other is through the government, which, in fact, is an intermediary agency. The government consists of up to m ($1 \leq m \leq 9$) employees conveniently numbered from 0 to $m-1$. Suppose employee z can introduce person x to person y at a cost of d dollars. If this is the first time in a day that employee z introduce one person to another, he will only require d dollars. For the second time, he will require d dollars plus extra e dollars as his tip. For the third time and more, he will require d dollars plus extra f dollars. He is not dared to require any more than that since the strange country is somewhat democratic. And if person x is able to communicate with person t and person t is able to communicate with person y , then person t is always willing to transfer messages from person x to person y , at no charge. Of course, the intermediary fees are all paid by person x . Notice that employee z being able to introduce person x to person y doesn't mean he can introduce person y to person x .

Now person 0 has to send a message to person $n-1$ in one day. If all employees have just started to work, what is the minimum cost for person 0 ?

Input

For each test case, the first line contains three integers, n , m and q , where q is the number of intermediary relationships and q is at most 10,000. The second line has m integers, each indicating the value e of every employee, in the range $[0, 100]$. The third line has m integers too, each indicating the value f of every employee, in the range $[e, 200]$. The next q lines each contains four integers, x , y , z and d , indicating that employee z can introduce person x to person y requiring d dollars, where $1 \leq d \leq 200$. There is a blank line after each test case.

Proceed to the end of file.

Output

For each test case, print one integer on a single line, giving the minimum cost. If it is impossible, print -1.

Samples

input

Copy

```
3 2 2
1 1
2 2
0 1 0 1
1 2 1 2
```

```
5 1 4
1
2
0 1 0 1
1 2 0 1
2 3 0 1
3 4 0 1
```

output [Copy](#)

```
3
9
```

Source

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Codes

HZNUOJ is based on [HUSTOJ](#)

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Server Time: 2025-8-26 14:58:50